

KANCHANA BURI, Thailand

WMO: 484500

Lat: 14.0225N

Lon: 99.5358E

Elev: 29

StdP: 100.98

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |            |             | MCWS/PCWD to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|------------|-------------|-----------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%         |             |                       |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS         | MCDB        | MCWS                  | PCWD      |              |
| (a)<br>12     | (b)<br>17.6 | (c)<br>19.0 | (d)<br>10.5                   | (e)<br>7.9 | (f)<br>24.8 | (g)<br>12.3 | (h)<br>8.9 | (i)<br>25.3 | (j)<br>5.4            | (k)<br>27.3 | (l)<br>4.6 | (m)<br>27.7 | (n)<br>0.1            | (o)<br>30 | (p)<br>0.332 |

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 4             | 10.3                   | 38.9            | 25.1 | 37.8 | 25.1 | 36.8 | 25.1 | 27.3                | 33.8 | 27.1 | 33.3 | 26.7 | 32.8 | 1.7                  | 150  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 26.0                            | 21.4 | 29.3 | 25.7 | 21.0 | 29.0 | 25.4 | 20.6 | 28.8 | 86.8          | 34.1 | 85.3 | 33.4 | 84.1 | 33.0 | 34.9              |

Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |      |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |     |
|-------------------|-----|------|----------------------------|------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|-----|
| 1%                |     | 2.5% | 5%                         | Mean |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |     |
| Min               |     | Max  |                            | Min  | Max  |                    | Min                                                | Max       | Min  | Max        | Min  | Max        | Min  | Max        | Min  | Max |
| (a)               | (b) | (c)  | DB                         | (e)  | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |     |
| 4.4               | 3.5 | 3.1  |                            | 15.4 | 40.5 | 2.2                | 1.3                                                | 13.9      | 41.5 | 12.6       | 42.2 | 11.4       | 42.9 | 9.8        | 43.9 |     |
|                   |     |      | WB                         | 13.1 | 28.5 | 2.0                | 1.5                                                | 11.7      | 29.6 | 10.5       | 30.4 | 9.3        | 31.3 | 7.9        | 32.4 |     |

Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 29.0   | 26.7  | 28.8  | 30.5  | 31.8  | 30.8  | 30.1  | 29.4  | 29.3  | 29.1  | 28.4  | 27.4  | 26.2  |
|                                                                                  | DBStd       | 2.38   | 2.12  | 2.06  | 2.07  | 1.89  | 1.95  | 1.39  | 1.39  | 1.26  | 1.28  | 1.30  | 1.72  | 2.28  |
|                                                                                  | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                                  | CDD10.0     | 6950   | 518   | 526   | 634   | 654   | 646   | 602   | 602   | 599   | 572   | 569   | 523   | 503   |
|                                                                                  | CDD18.3     | 3908   | 260   | 293   | 376   | 404   | 388   | 352   | 344   | 341   | 322   | 311   | 273   | 245   |
|                                                                                  | CDH23.3     | 45993  | 2674  | 3461  | 4867  | 5518  | 4996  | 4330  | 4041  | 3932  | 3578  | 3384  | 2822  | 2390  |
| CDH26.7                                                                          | 21748       | 1197   | 1759  | 2649  | 3221  | 2625  | 2028  | 1750  | 1653  | 1470  | 1321  | 1113  | 963   |       |
| Wind                                                                             | WSAvg       | 0.6    | 0.5   | 0.6   | 0.7   | 0.7   | 0.6   | 0.6   | 0.6   | 0.7   | 0.6   | 0.5   | 0.6   | 0.6   |
| Precipitation                                                                    | PrecAvg     | 1095   | 7     | 17    | 37    | 81    | 166   | 82    | 93    | 104   | 223   | 209   | 63    | 14    |
|                                                                                  | PrecMax     | 1581   | 78    | 90    | 213   | 146   | 286   | 179   | 193   | 199   | 414   | 367   | 196   | 76    |
|                                                                                  | PrecMin     | 857    | 0     | 0     | 0     | 6     | 58    | 33    | 29    | 10    | 59    | 89    | 0     | 0     |
|                                                                                  | PrecStd     | 221    | 20    | 30    | 52    | 48    | 51    | 39    | 40    | 55    | 93    | 71    | 51    | 22    |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 35.8  | 37.8  | 39.5  | 40.9  | 40.0  | 37.7  | 36.3  | 36.2  | 35.8  | 34.7  | 35.0  | 35.0  |
|                                                                                  |             | MCWB   | 23.4  | 23.2  | 23.3  | 25.3  | 26.4  | 25.9  | 25.9  | 26.0  | 25.9  | 26.4  | 25.4  | 24.5  |
|                                                                                  | 2%          | DB     | 34.5  | 36.7  | 38.1  | 39.2  | 38.0  | 36.0  | 35.0  | 34.8  | 34.6  | 33.6  | 33.7  | 33.7  |
|                                                                                  |             | MCWB   | 23.3  | 23.7  | 23.9  | 25.4  | 26.2  | 26.1  | 25.9  | 25.9  | 26.0  | 26.2  | 25.2  | 24.1  |
|                                                                                  | 5%          | DB     | 33.5  | 35.7  | 37.1  | 38.0  | 36.5  | 34.7  | 33.9  | 33.7  | 33.6  | 32.9  | 32.7  | 32.5  |
|                                                                                  |             | MCWB   | 23.2  | 23.8  | 24.3  | 25.6  | 26.2  | 26.1  | 25.6  | 25.8  | 25.8  | 25.9  | 24.7  | 23.6  |
|                                                                                  | 10%         | DB     | 32.3  | 34.5  | 36.0  | 36.8  | 35.1  | 33.6  | 32.9  | 32.7  | 32.6  | 32.0  | 31.7  | 31.4  |
|                                                                                  |             | MCWB   | 22.8  | 23.8  | 24.4  | 25.5  | 26.2  | 25.9  | 25.5  | 25.6  | 25.7  | 25.7  | 24.3  | 23.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 25.7  | 26.3  | 27.0  | 27.7  | 27.9  | 27.5  | 27.1  | 27.1  | 27.2  | 27.2  | 27.0  | 26.2  |
|                                                                                  |             | MCDB   | 32.0  | 34.1  | 34.3  | 35.9  | 35.5  | 33.7  | 33.0  | 32.8  | 32.5  | 32.0  | 32.3  | 31.9  |
|                                                                                  | 2%          | WB     | 25.1  | 25.7  | 26.3  | 27.2  | 27.2  | 27.0  | 26.6  | 26.6  | 26.7  | 26.9  | 26.4  | 25.3  |
|                                                                                  |             | MCDB   | 31.2  | 33.0  | 33.2  | 34.7  | 34.3  | 33.0  | 32.3  | 32.1  | 31.9  | 31.6  | 31.2  | 30.9  |
|                                                                                  | 5%          | WB     | 24.6  | 25.3  | 25.9  | 26.7  | 26.9  | 26.6  | 26.2  | 26.2  | 26.4  | 26.6  | 26.0  | 24.7  |
|                                                                                  |             | MCDB   | 30.3  | 32.0  | 32.8  | 34.0  | 33.6  | 32.3  | 31.7  | 31.7  | 31.5  | 31.1  | 30.4  | 30.2  |
|                                                                                  | 10%         | WB     | 24.0  | 24.9  | 25.6  | 26.4  | 26.5  | 26.2  | 25.9  | 25.9  | 26.1  | 26.2  | 25.5  | 24.1  |
|                                                                                  |             | MCDB   | 29.4  | 31.1  | 32.2  | 33.5  | 32.9  | 31.6  | 31.2  | 31.1  | 31.0  | 30.5  | 29.6  | 29.4  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 11.2  | 11.5  | 10.8  | 10.3  | 8.4   | 7.5   | 7.2   | 7.2   | 7.4   | 7.1   | 8.1   | 9.8   |
|                                                                                  |             | MCDBR  | 12.3  | 12.6  | 12.3  | 11.6  | 10.1  | 8.9   | 8.6   | 8.5   | 8.7   | 8.2   | 9.4   | 10.7  |
|                                                                                  |             | MCWBR  | 3.6   | 3.3   | 3.3   | 2.7   | 2.4   | 2.3   | 2.3   | 2.3   | 2.4   | 2.4   | 2.8   | 3.1   |
|                                                                                  | 5% WB       | MCDBR  | 10.2  | 11.0  | 10.7  | 10.4  | 8.9   | 7.9   | 7.6   | 7.7   | 7.8   | 7.5   | 8.1   | 9.6   |
| MCWBR                                                                            |             | 3.3    | 3.0   | 3.0   | 2.8   | 2.5   | 2.3   | 2.4   | 2.4   | 2.5   | 2.6   | 2.7   | 2.9   |       |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        |        | 0.536 | 0.602 | 0.647 | 0.565 | 0.481 | 0.444 | 0.439 | 0.439 | 0.459 | 0.573 | 0.525 | 0.516 |
|                                                                                  | taud        |        | 1.956 | 1.824 | 1.748 | 1.991 | 2.279 | 2.385 | 2.399 | 2.385 | 2.316 | 1.950 | 2.057 | 2.025 |
|                                                                                  | Ebn at Noon |        | 762   | 730   | 710   | 770   | 823   | 845   | 852   | 861   | 849   | 744   | 767   | 767   |
|                                                                                  | Edh at Noon |        | 181   | 214   | 236   | 185   | 136   | 121   | 120   | 123   | 132   | 186   | 162   | 165   |
| All-Sky Solar<br>Radiation                                                       | RadAvg      |        | 4.75  | 5.24  | 5.47  | 5.96  | 5.32  | 4.85  | 4.53  | 4.51  | 4.72  | 4.48  | 4.64  | 4.62  |
|                                                                                  | RadStd      |        | 0.29  | 0.24  | 0.43  | 0.36  | 0.47  | 0.40  | 0.38  | 0.34  | 0.35  | 0.31  | 0.33  | 0.22  |

Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                         |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| Station Only            | N/A   | N/A     | N/A    | N/A     | +0.19 | +0.24 | N/A         | N/A     | N/A     | N/A     |
| Regional (23 neighbors) | +0.21 | N/A     | N/A    | N/A     | +0.26 | +0.28 | N/A         | N/A     | +100    | +100    |

Nomenclature: See separate page